Sparse parity under globally conditioned neural priors

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Abstract

We study posterior confidence for a parity of k coordinates in a d-dimensional Boolean input under a specified Bayesian neural-network prior. The network has fixed depth, Gaussian weights, hyperbolic-tangent hidden activations, and a linear readout. Its prior is conditioned to have a fixed absolute output margin on every input, followed by exact training labels. For every fixed negative width-scaling exponent $\gamma < 0$, we prove that each sample size $n \le \exp\{ck \log(ed/k)\}$ leaves a universal positive fraction of inputs without correct-label posterior probability greater than $2/3$, with probability at least $3/4$, in the width-first limit. This is a $d^{\Omega(k)}$ lower bound when $k \le d^{1-\eta}$ for fixed $\eta > 0$. The proof uses Gaussian sign conditioning, sharp Fourier spectra, near-orthogonal covariance bounds, and a uniform truncated negative-part estimate. The distinction between scaling regimes is substantive: for every $0 < \gamma < 1$, the same prior learns a one-coordinate parity with $O(\log d)$ samples. We prove this by identifying its exact minimum energy sign functions. These results do not settle the corresponding growing-k lower bound at $\gamma = 0$ or $0 < \gamma < 1$.

1 Model and results

Let $\mathcal{X} = \{-1, 1\}^d$ with uniform probability measure. Fix $U \subseteq [d]$, $|U| = k$, and set $f(x) = \chi_U(x) = \prod_{j \in U} x_j$. All hidden layers have width N, the number of weight matrices $L \ge 2$ is fixed, and

$$h^0(x) = (1, x), \qquad h^s(x) = \tanh(W^s h^{s-1}(x)) \quad (1 \le s < L), \qquad z_N(x) = W^L h^{L-1}(x).$$

There are no additional biases. All weights are independent centered Gaussians, with variances

$$\operatorname{Var} W^1_{ij} = N^{1-\gamma}/d, \qquad \operatorname{Var} W^s_{ij} = N^{-\gamma} \quad (1 < s < L), \qquad \operatorname{Var} W^L_i = N^{-\gamma-1}. \tag{1}$$

Fix $a > 0$; in particular, one may take $a = 1/2$. Condition the prior on $|z_N(x)| > a$ for every $x \in \mathcal{X}$. For independent uniform training inputs $S = (X_1, \dots, X_n)$, condition subsequently on $f(X_i) z_N(X_i) > 0$ for every i. Equivalently, condition on the intersection of these events. Define

$$q_{N,S}(x) = \mathbb{P}\{f(x) z_N(x) > 0 \mid |z_N(y)| > a \ \forall y;\ f(X_i) z_N(X_i) > 0 \ \forall i\},$$
$$F_N(S) = \mathbb{P}_x\{q_{N,S}(x) \le 2/3\}.$$

Thus a successful prediction requires strictly more than $2/3$ posterior probability of the correct label. A fresh input is always independent and uniform. The learner is not told the parity support.

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Theorem 1.1 (Sparse-parity sample lower bound)**.** *For each fixed $L \geq 2$ and $a > 0$, there are $c > 0$ and d_0 such that, for every $d \geq d_0$, every $1 \leq k \leq d$, every support U of size k, every integer*

$$0 \leq n \leq \exp\left(ck \log \frac{ed}{k}\right),$$

and every fixed $\gamma < 0$,

$$\mathbb{P}_S \left\{ \liminf_{N\to\infty} F_N(S) \geq \varepsilon \right\} \geq \frac{3}{4}, \qquad \varepsilon = \frac{2\pi(1/e - 7/20)^2}{256} > 0.$$

The conditioning events have positive probability for all sufficiently large widths at each fixed dimension and sample.

The numerical value of the displayed failure fraction is approximately 7.85×10^{-6}. It is uniform in d, k, L, a, although the sample exponent and dimension threshold are not quantitatively optimized. In particular, the theorem rules out vanishing confidence failure below the stated sample scale; it does not assert a failure fraction as large as $1/10$. For any fixed $\eta \in (0, 1)$ and $k \leq d^{1-\eta}$, the bound includes every $n \leq d^{c\eta k}$. When $k = d$ it becomes $n \leq e^{cd}$. The latter dependence is necessary for a uniform statement: the input space has only 2^d points, and observing almost all of them suffices for this confidence criterion under noiseless conditioning.

The width limit is taken first, at fixed d, k, L, a, γ, S. No simultaneous dimension–width limit or quantified minimum width is asserted. The probability bound holds for each specified sample size with uniform constants, rather than on a common event for all prefixes of an infinite sample sequence. It uses no monotonicity assumption on fixed-target posterior risk.

Theorem 1.2 (A separation between scaling regimes)**.** *Fix $d \geq 1$, $L \geq 2$, $a > 0$, and $0 < \delta < 1$. For $0 < \gamma < 1$, a target $f(x) = x_1$ and*

$$n \geq \left\lceil \log_2 \frac{2d}{\delta} \right\rceil$$

satisfy $q_{N,S}(x) \to 1$ for every input, with probability at least $1 - \delta$ over training samples. In contrast, for $\gamma < 0$, every target and every sample of size n satisfy

$$\liminf_{N\to\infty} F_N(S) \geq 1 - \left(2 + \frac{400}{9\pi}\right) \frac{n}{d+1}.$$

Thus a d^{ck} lower bound cannot hold uniformly over all $\gamma < 1$ and all $k \geq 1$. Theorem 1.2 does not resolve the lower-bound question restricted to growing k for $0 \leq \gamma < 1$.

1.1 Proof overview

Put $\Psi(t) = 2\arcsin(t)/\pi$ and $\beta = 2^{-(L-1)}$. The normalized network outputs converge to a Gaussian vector with covariance

$$K(x, y) = \Psi^{\circ(L-1)}\left(\frac{1 + x \cdot y}{d+1}\right).$$

We regard K as a 2^d-by-2^d covariance matrix. Its normalized operator is $2^{-d}K$, and the kernel norm of a field h is $\|h\|_K^2 = h^\mathsf{T} K^{-1} h$. Population norms, in contrast, use the uniform probability measure on the cube. Keeping these two normalizations separate is essential in the spectral and sample bounds. For $\gamma < 0$, the fixed global margin vanishes after normalization, leaving Gaussian conditioning on the training signs. A homogeneous-cone argument bounds the energy of its actual

posterior mean by n. Log-concavity and Gaussian regression convert posterior confidence into a quantitative mean margin wherever residual variance remains. These are implications about the actual posterior, not a replacement of its predictions by the sign of its mean.

We split that mean into a low-degree part, one further pair of Fourier levels, and a remaining tail. Uniform sampling controls a truncated negative part of the first two pieces. Orthogonality to the target parity forces their positive and negative integrals to agree whenever their degree is below k. Sharp covariance estimates control the tail and prevent the negative integral from being hidden on a tiny set. A near-orthogonal entry bound is what permits cutoffs much smaller than $\log d$. This yields the ambient-dimension exponent uniformly in k.

For $0 < \gamma < 1$, global-margin conditioning instead selects compatible sign functions of minimum Gaussian energy. The top eigenspace of the kernel consists exactly of affine functions, and the only globally constant-magnitude functions in it are constants and signed coordinates. This identifies the ground states and proves the logarithmic upper bound. All required width-limit, spectral, sampling, and symmetry arguments are included below.

2 The kernel rate of the actual network

Let $M = 2^d$, and identify output functions with vectors in $\mathbb{R}^M$. Define

$$\Psi(t) = \frac{2}{\pi} \arcsin t, \qquad K(x,y) = \Psi^{\circ(L-1)}\left(\frac{1 + x \cdot y}{d+1}\right). \tag{2}$$

The positive definiteness of K is proved in the next section. For each Boolean function g, put

$$E_K(g) = \min_{g(x)z(x) \geq a \ \forall x} \frac{1}{2} z^T K^{-1} z. \tag{3}$$

The minimum exists since its feasible set is closed and nonempty and the objective is coercive. Replacing the closed inequalities by strict ones gives the same infimum, by scaling a feasible vector by $1 + \epsilon$.

Proposition 2.1 (Posterior concentration on minimum-energy signs)**.** *For the actual network* (1)*, at every fixed* $0 < \gamma < 1$*,*

$$\lim_{N \to \infty} N^{-\gamma} \log \mathbb{P}\{g(x) z_N(x) > a \ \forall x\} = -E_K(g). \tag{4}$$

Consequently, after any fixed noiseless training sample, posterior mass tends to one on the compatible functions minimizing E_K*. No assertion about the relative weights of tied minimizers is needed.*

Proof. For hidden layer s, let $Q_N^s(x,y) = N^{-1} \sum_i h_i^s(x) h_i^s(y)$. We first establish, for each fixed $\epsilon > 0$,

$$\mathbb{P}\{\|Q_N^{L-1} - K\| > \epsilon\} \leq C_\epsilon e^{-c_\epsilon N} \tag{5}$$

for all sufficiently large N. Any matrix norm can be used, since M is fixed. Conditional on the preceding layer, the rows of the next hidden layer are independent bounded vectors. Hoeffding's inequality controls all their empirical second moments. Their preactivations have covariance $N^{1-\gamma} Q_N^{s-1}$; at the first layer the covariance is $N^{1-\gamma}(1 + x \cdot y)/d$. Because $N^{1-\gamma} \to \infty$, the conditional second-moment map converges to the Gaussian-sign map

$$Q \longmapsto \left[\Psi\left(\frac{Q(x,y)}{\sqrt{Q(x,x)Q(y,y)}}\right)\right]_{x,y}.$$

This follows by dominated convergence for a Gaussian vector, since each scalar variance stays bounded away from zero in a neighborhood of the limiting covariance. The convergence is uniform in such a neighborhood: a compactness subsequence and a continuous square root coupling reduce the assertion to the same dominated-convergence argument. The limiting map is continuous there. Inducting through the fixed number of layers, with smaller fixed tolerances at earlier layers, proves (5). The first-layer limit normalizes the diagonal $(d+1)/d$, giving the kernel in (2).

Conditionally on the hidden layers, the readout is exactly Gaussian:

$$Z_N \mid Q_N^{L-1} \sim \mathcal{N}(0, N^{-\gamma} Q_N^{L-1}). \tag{6}$$

Couple it to $Y_N = N^{-\gamma/2} K^{1/2} \xi$, where $\xi \sim \mathcal{N}(0, I_M)$ is independent of the hidden layers. On a fixed sufficiently small neighborhood of the positive-definite matrix K, the square-root map is Lipschitz. Hence, for $\delta > 0$,

$$\mathbb{P}\{\|Z_N - Y_N\| > \delta\} \le C_\epsilon e^{-c_\epsilon N} + \mathbb{P}\{\|\xi\| > \delta N^{\gamma/2}/(C\epsilon)\}.$$

First divide logarithms by N^γ and let $N \to \infty$; then let $\epsilon \downarrow 0$. The resulting upper limit is $-\infty$, since $\gamma < 1$. The coupled vectors are therefore exponentially equivalent at speed N^γ.

The fixed-dimensional Gaussian vectors Y_N have large-deviation rate $I(z) = z^T K^{-1} z/2$, directly from their densities. Exponential equivalence transfers the same bounds to Z_N. Applying the upper bound to a closed margin orthant and the lower bound to its interior gives (4), since their infima agree. Finally, there are only finitely many Boolean functions for fixed d. Dividing each compatible orthant probability by their sum proves the posterior concentration claim. □

This is a rate statement for the network itself. It does not require relative-error approximation of rare probabilities, nor convergence in total variation to a Gaussian posterior.

Proposition 2.2 (The nonpositive-exponent posterior limits)**.** *For $\gamma = 0$, the Boolean posterior converges to that of $G \sim \mathcal{N}(0, K)$ conditioned on $|G(x)| > a$ everywhere and on the observed signs. For $\gamma < 0$, it converges to the Gaussian posterior conditioned only on the observed signs.*

Proof. The hidden-kernel argument establishing (5) holds for every $\gamma < 1$. By (6), the normalized vector $N^{\gamma/2} Z_N$ therefore converges in law to $\mathcal{N}(0, K)$. Its all-input margin is $aN^{\gamma/2}$, which equals a when $\gamma = 0$ and tends to zero when $\gamma < 0$. The limiting Gaussian is nonsingular. Thus all boundaries involved in any fixed full-sign event or margin event have probability zero, and the conditioning events have positive probability. Taking ratios of their probabilities gives convergence of each Boolean posterior mass. There are finitely many masses for fixed d. □

3 Fourier level masses of the kernel

Put $D = d + 1$, $\beta = 2^{-(L-1)}$, and $F = \Psi^{\circ(L-1)}$. The characters $\chi_U(x) = \prod_{j \in U} x_j$ diagonalize the normalized covariance operator $M^{-1}K$, $M = 2^d$. Let λ_j be its eigenvalue at degree j. We first work on the full D-cube with kernel $F(u \cdot v/D)$, whose degree-j eigenvalue is denoted by a_j. Constants in this and the next section may depend on L. Notation $b \asymp c$ means two inequalities by positive constant multiples, uniformly in the stated ranges.

Lemma 3.1 (Power coefficients and paired eigenvalues)**.** *Write $F(z) = \sum_{m \text{ odd}} c_m z^m$. Every odd coefficient is positive, $\sum_m c_m = 1$, and*

$$c_m \asymp \frac{\Gamma(m - \beta)}{m!} \asymp m^{-1-\beta} \qquad (m \ge 1 \text{ odd}).$$

The full-cube eigenvalues vanish at even degrees and strictly decrease through odd degrees. They are positive at every odd degree. On the original d-cube,

$$\lambda_{2k} = \lambda_{2k+1} = a_{2k+1} > 0,$$

with a final unpaired eigenvalue if d is even. In particular K is positive definite.

Proof. The series of Ψ has positive coefficients at every odd exponent, with

$$[z^{2k+1}]\Psi(z) = \frac{2}{\pi}\frac{\binom{2k}{k}}{4^k(2k+1)}.$$

Composition preserves strict positivity and oddness. Near $z = 1$ in a slit neighborhood,

$$\Psi(z) = 1 - b(1-z)^{1/2} + O((1-z)^{3/2}), \qquad b = \frac{2\sqrt{2}}{\pi}.$$

Induction gives $F(z) = 1 - A(1-z)^{\beta} + O((1-z)^{3\beta})$, with $A > 0$; at each composition its leading constant changes from A_t to $b\sqrt{A_t}$, starting from $A_1 = b$. These expansions admit algebraic coefficient transfer [2, Chapter VI]. Here are the analytic-domain details. On the unit circle outside $\{-1, 1\}$, positivity of every odd coefficient gives $|\Psi(z)| < 1$: equality in the triangle inequality would require $z^2 = 1$. Hence the iterated functions extend analytically through every compact arc away from these endpoints. At 1, restrict $\arg(1-z)$ to a closed proper subinterval of $(-\pi, \pi)$. Each square-root composition reduces this angular range, and its positive leading coefficient keeps its image in the slit neighborhood for the next arcsine composition. Oddness gives the identical argument at -1. These local sectors and the compact arcs provide the required two-singularity indented domain. The contributions of 1 and -1 add at odd exponents, giving

$$c_m \sim \frac{2A}{-\Gamma(-\beta)} m^{-1-\beta} \quad (m \to \infty \text{ odd}).$$

Together with positivity of the finitely many initial coefficients and the gamma-ratio asymptotic, this proves the coefficient comparisons. For $L = 2$ the explicit formula and Stirling's bounds suffice instead.

Let $p_m(j)$ be the probability that the simple coordinate-flip walk on the D-cube reaches a specified distance-j vertex in m steps. Expanding $(u \cdot v/D)^m$ shows that its degree-j Fourier coefficient is $p_m(j)$. Thus $a_j = \sum_{m \text{ odd}} c_m p_m(j)$. For the walk recursion,

$$\begin{aligned} D[p_{m+1}(j) - p_{m+1}(j+2)] = j[p_m(j-1) - p_m(j+1)] \\ + (D-j-2)[p_m(j+1) - p_m(j+3)]. \end{aligned}$$

Induction from time zero gives nonincreasing probabilities through each parity class; terms with zero boundary coefficients are omitted. The term $m = j$ makes the odd-eigenvalue inequalities strict and proves positivity. Even-degree eigenvalues vanish by parity. Restricting the distinguished coordinate to $+1$ gives $\lambda_q = a_q + a_{q+1}$, proving the claims. □

For odd r, the exact combined multiplicity of reduced degrees $j-1, j$ is $\binom{d}{j-1} + \binom{d}{j} = \binom{D}{j}$. Consequently

$$D_r = \sum_{\substack{j \le r \\ j \text{ odd}}} \binom{D}{j}, \quad \tau_r = \sum_{\substack{j > r \\ j \text{ odd}}} \binom{D}{j} a_j, \quad \lambda_{r+1} = a_{r+2}. \tag{7}$$

The odd cutoff is essential for these exact identities.

Lemma 3.2 (Sharp level and tail estimates)**.** *Uniformly for odd* $1 \le j \le D/4$ *and odd* $1 \le r \le D/8$,

$$\binom{D}{j} a_j \asymp j^{-1-\beta}, \tag{8}$$

$$\tau_r \asymp r^{-\beta}, \tag{9}$$

$$\frac{\lambda_{r+1}}{\tau_r} \asymp \frac{1}{r\binom{D}{r+2}}, \tag{10}$$

$$s_{2,r} := \sum_{q>r} \binom{d}{q} \lambda_q^2 \le \frac{C r^{-2\beta-2}}{\binom{D}{r+2}}. \tag{11}$$

The same level-mass estimate holds up to $j \le \rho D$ *for any fixed* $\rho < 1/2$*, with constants depending on* ρ*.*

Proof. The gamma integral and Lemma 3.1 give, for odd j,

$$a_j \asymp D^{-\beta} \int_0^\infty u^{-1-\beta} p(u)^j (1-p(u))^{D-j}\, du, \quad p(u) = \frac{1-e^{-2u}}{2}, \tag{12}$$

$$\asymp D^{-\beta} \int_0^{1/2} p^j (1-p)^{D-j} w_\beta(p)\, dp, \tag{13}$$

where $w_\beta(p) = [-\frac{1}{2}\log(1-2p)]^{-1-\beta}/(1-2p)$. Indeed the gamma integral Poissonizes the walk time, and independent Poisson coordinate counts give independent coordinate parities with success probability $p(u)$. Since an odd-distance vertex has zero probability at even walk times, no extra parity factor occurs. The weight w_β is integrable near $1/2$ and comparable to $p^{-1-\beta}$ on $[0, b]$ for every fixed $b < 1/2$.

Fix $\rho < b < 1/2$ and take $j \le \rho D$. The integral on $[0, b]$ is comparable to the complete beta integral $B(j-\beta, D-j+1)$. For the lower comparison, a beta random variable with these parameters has mean at most ρ, so Markov's inequality puts at least a fraction $1 - \rho/b$ of its mass in $[0, b]$. The identity

$$\binom{D}{j} B(j-\beta, D-j+1) = \frac{\Gamma(D+1)\Gamma(j-\beta)}{\Gamma(j+1)\Gamma(D+1-\beta)} \asymp D^\beta j^{-1-\beta}$$

gives the desired mass. On $[b, 1/2]$, the binomial factor decreases with p and, after multiplication by $\binom{D}{j}$, is at most $\exp[-D\,\mathrm{KL}(\rho\|b)]$. The remaining integrable weight makes this contribution exponentially small. This proves (8) and its stated extended range.

Sum (12) over the odd distances above r. Dropping the parity restriction and using the binomial Markov bound gives

$$\tau_r \le C D^{-\beta} \int_0^\infty u^{-1-\beta} \min\{1, Du/(r+1)\}\, du \le C(r+1)^{-\beta}.$$

For the lower bound, sum (8) over the odd j in $(r, 2r+2]$, using a fixed cutoff strictly below $D/2$ as needed at the endpoint. This yields $cr^{-\beta}$; bounded r are absorbed by a change of constants. Equation (10) follows from (8), (9), and (7).

For the second moment, use paired multiplicities to write $s_{2,r} = \sum_{j>r,\ j \text{ odd}} \binom{D}{j} a_j^2$. Up to an odd cutoff b_D near $D/4$, its terms are at most $Cj^{-2\beta-2}/\binom{D}{j}$. Consecutive odd inverse binomial coefficients have ratio

$$\frac{(j+1)(j+2)}{(D-j)(D-j-1)},$$

uniformly less than a fixed constant below one in this range. The sum is therefore bounded by a constant times its first term, at $j = r + 2$. Beyond b_D, monotonicity gives the bound $a_{b_D}\tau_{b_D} \le CD^{-2\beta-1}/\binom{D}{b_D}$, which is exponentially smaller than the claimed bound uniformly for $r \le D/8$. This proves (11). □

Finally, two-step binomial ratios give, uniformly for odd $r \le D/8$,

$$\binom{D}{r} \le D_r \le C\binom{D}{r}, \qquad \frac{D_r}{\binom{D}{r+2}} \le C(r/D)^2. \tag{14}$$

All constants used here can be fixed independently of a subsequently chosen small positive limit $r/d \to \alpha$.

4 Dictators separate the subcritical regimes

The smallest nonconstant parity already shows that a lower bound d^{ck} cannot hold uniformly over all $k \ge 1$ throughout $\gamma < 1$. Here and below the width tends to infinity at fixed d, training sample, and exponent. We use the normalized Walsh eigenvalues λ_j of $K/2^d$.

Lemma 4.1 (Exact ground states)**.** *The largest normalized eigenvalue is $\lambda_* := \lambda_0 = \lambda_1$, and its eigenspace consists exactly of affine functions on the cube. Moreover, $\lambda_* \le 1/(d+1)$. Among all vectors with $|z(x)| \ge a$ at every input, the minimum of $z^{\mathsf{T}}K^{-1}z/2$ is $a^2/(2\lambda_*)$. Its minimizers are precisely*

$$z \in \{\pm a,\ \pm ax_1, \dots, \pm ax_d\}.$$

Proof. The pairing and strict spectral ordering follow from Lemma 3.1. For an elementary verification of strictness, put $D = d + 1$ and write $F(t) = \sum_{m \text{ odd}} c_m t^m$, where every odd coefficient is positive. The degree-j eigenvalue on the full D-cube is $a_j = \sum_m c_m p_m(j)$, with $p_m(j)$ the coordinate-flip walk probability of a specified distance-j vertex. Its exponential generating formula is

$$p_m(j) = \frac{m!}{D^m}[t^m](\sinh t)^j(\cosh t)^{D-j}.$$

Subtracting the formula at $j + 2$ leaves $(\sinh t)^j(\cosh t)^{D-j-2}$, whose coefficients are nonnegative. The coefficient at $m = j$ is positive. Thus $a_j > a_{j+2}$ at every available odd pair, and restricting the distinguished coordinate gives $\lambda_{2j} = \lambda_{2j+1}$. Consequently the top eigenspace on the original cube has dimension $d+1$ and consists of degrees zero and one. The normalized covariance has trace $K(x,x) = 1$, so $(d+1)\lambda_* \le 1$.

For the uniform population norm,

$$z^{\mathsf{T}}K^{-1}z = \sum_U \frac{\widehat{z}(U)^2}{\lambda_{|U|}} \ge \frac{\|z\|_2^2}{\lambda_*} \ge \frac{a^2}{\lambda_*}.$$

Equality requires both $|z(x)| = a$ everywhere and membership in the top eigenspace. If $P = z/a = c_0 + \sum_j c_j x_j$ takes values in $\{-1, 1\}$, then $P^2 = 1$. Its degree-two coefficients give $c_i c_j = 0$ for $i \ne j$, and its degree-one coefficients give $c_0 c_i = 0$. Hence at most one of $c_0, c_1, \dots, c_d$ is nonzero; the constant coefficient of P^2 makes that coefficient ± 1. The listed functions attain equality. □

Theorem 4.2 (Logarithmic samples for a dictator)**.** *Fix $L \geq 2$, $a > 0$, $0 < \gamma < 1$, and $0 < \delta < 1$. For target $f(x) = x_1$, let S consist of independent uniform training inputs. If*

$$n \geq \left\lceil \log_2 \frac{2d}{\delta} \right\rceil,$$

then, with probability at least $1 - \delta$,

$$\lim_{N\to\infty} q_{N,S}(x) = 1 \qquad \text{for every } x \in \mathcal{X}.$$

In particular the strict-2/3 confidence-failure fraction is zero for all sufficiently large widths on this event.

Proof. The target is always a compatible ground-state sign. Its opposite $-x_1$ is excluded by any training observation. Each of the other $2d$ constants or signed coordinate functions agrees with the target at a uniform input with probability $1/2$, and hence on all n independent inputs with probability 2^{-n}. A union bound shows that the target is the unique compatible ground-state sign with probability at least $1 - 2d\,2^{-n} \geq 1 - \delta$.

On this event, Lemma 4.1 makes the target the unique compatible sign function attaining the global minimum energy. Proposition 2.1 therefore puts posterior mass tending to one on this sign function. This proves the assertion simultaneously at all inputs, since the cube is finite at each width limit. No formula for weights among tied minimizers is used. □

The contrast with the negative-exponent limit concerns actual posterior probabilities as well. It does not identify confidence with the sign of a Gaussian posterior mean.

Theorem 4.3 (A universal linear-dimensional obstruction for negative exponents)**.** *Let $f : \mathcal{X} \to \{-1, 1\}$ be any target and S any training sample of size n, allowing repetitions. Under the Gaussian prior conditioned on the observed signs, set*

$$F_\theta(S) = \mathbb{P}_x\{\mathbb{P}(f(x)G(x) > 0 \mid f(X_i)G(X_i) > 0\ \forall i) \leq \theta\}.$$

For every $1/2 < \theta < 1$,

$$F_\theta(S) \geq 1 - \left(2 + \frac{1}{\pi(\theta - 1/2)^2}\right) \frac{n}{d+1}. \tag{15}$$

Consequently, for every fixed $\gamma < 0$, the actual neural-network posterior with the stipulated global-margin convention satisfies

$$\liminf_{N\to\infty} F_N(S) \geq 1 - \left(2 + \frac{400}{9\pi}\right) \frac{n}{d+1}. \tag{16}$$

In particular $n = o(d)$ leaves confidence failure $1 - o(1)$, even for a dictator or a constant target.

Proof. Remove repeated training inputs and let $s \leq n$ be their number. Write Gaussian regression on these values as

$$G(x) = R(x) + \varepsilon(x), \qquad R = K_{\mathcal{X},S} K_{SS}^{-1} G_S, \qquad \sigma_x^2 = \operatorname{Var} \varepsilon(x).$$

The residual process is independent of G_S. Whiten the training vector by $G_S = K_{SS}^{1/2}\xi$. The observed signs form a homogeneous cone in the standard Gaussian vector ξ. Its radius is independent of its

direction, so conditioning on this cone preserves $\mathbb{E}\|\xi\|^2 = s$. With $\mathbb{E}_S^*$ denoting expectation under that sign conditioning, the regression identity gives

$$\|R\|_K^2 = \|\xi\|^2, \qquad \mathbb{E}_S^*\|R\|_2^2 \le \lambda_* s \le \lambda_* n.$$

These statements concern the second moment of the random regression field, not just its posterior mean.

The prior covariance of R is positive semidefinite, bounded by K, and has rank at most s. Its normalized trace is at most $s\lambda_*$. Since $\sigma_x^2 = 1 - \operatorname{Var} R(x)$,

$$\mathbb{P}_x\{\sigma_x^2 < 1/2\} \le 2n\lambda_*.$$

For every remaining input, conditional Gaussian regression yields

$$q_S(x) = \mathbb{E}_S^*\Phi\left(\frac{f(x)R(x)}{\sigma_x}\right),$$

where Φ is the standard normal distribution function. The inequality $|\Phi(t) - 1/2| \le |t|/\sqrt{2\pi}$ and $\sigma_x \ge 1/\sqrt{2}$ imply

$$|q_S(x) - 1/2| \le \frac{\sqrt{\mathbb{E}_S^* R(x)^2}}{\sqrt{\pi}}.$$

Thus $q_S(x) > \theta$ at such an input requires $\mathbb{E}_S^* R(x)^2 > \pi(\theta - 1/2)^2$. Markov's inequality, integrated over inputs, bounds the fraction of these inputs by $n\lambda_*/[\pi(\theta - 1/2)^2]$. Adding the variance exception and using $\lambda_* \le 1/(d+1)$ proves (15).

Take $\theta = 13/20 < 2/3$. For $\gamma < 0$, Proposition 2.2 gives convergence of each of the finitely many correct-label posterior probabilities to the sign-conditioned Gaussian value. Every input with limiting value at most $13/20$ therefore has finite-width value below $2/3$ for all sufficiently large widths. This proves (16). The case $n = 0$ follows directly from sign symmetry and also satisfies the displayed bounds. □

5 An exact pairing of parity sizes

The following symmetry holds before taking any width limit.

Proposition 5.1 (Projective-cube symmetry)**.** *At every width and exponent for which the conditioning is defined, the iid-sample learning laws of coordinate parities of sizes* 0 *and* 1*, of sizes* 2 *and* 3*, and so on, coincide whenever both sizes are available in ambient dimension* d*. The equality includes the distribution of the actual confidence-failure fraction.*

Proof. Fix a coordinate i and define an involution of the cube by

$$(T_i x)_i = x_i, \qquad (T_i x)_j = x_i x_j \quad (j \ne i).$$

If P_i swaps augmented coordinates 0 and i, then $(1, T_i x) = x_i P_i(1, x)$. For fixed weights W, replace only the first matrix by $W'^1 = W^1 P_i$. Oddness of every hidden activation and of the readout gives, inductively,

$$z_W(T_i x) = x_i z_{W'}(x).$$

The laws of W and W' agree because the first-layer columns are exchangeable. For a target g, define $g'(T_i x) = x_i g(x)$. Absolute margins are invariant under this transformation, and the training-label

event for $(g', T_i S)$ becomes exactly the event for (g, S). Thus the correct-label posterior probabilities and their failure fractions agree after the change of variables. The involution preserves the uniform input distribution.

For $g = \chi_U$, $|U| = k$, writing $u = T_i x$ gives

$$g'(u) = u_i^{k+1} \prod_{j \in U \setminus \{i\}} u_j.$$

If k is odd and $i \in U$, this is a $(k-1)$-coordinate parity. If k is even and $i \notin U$, it is a $(k+1)$-coordinate parity. Ordinary coordinate permutations identify all supports of a fixed size, proving the assertion. □

These results refute a d^{ck} lower bound asserted uniformly including $k = 1$ across the entire subcritical regime. They do not refute a lower bound restricted to $k \to \infty$, and the symmetry reduces parity size only within one adjacent even–odd pair.

6 The actual Gaussian sign posterior

Let $\mathcal{X} = \{-1, 1\}^d$, $M = 2^d$, and $f(x) = \prod_{j \in U} x_j$, $\quad |U| = k$. All function norms and expectations on $\mathcal{X}$ use its uniform probability measure. Let G be a nondegenerate centered Gaussian vector indexed by $\mathcal{X}$, with covariance matrix K. The training inputs $X_1, \dots, X_n$ are independent and uniform. We condition on

$$\mathcal{A}_S = \{f(X_i)G(X_i) > 0 : 1 \le i \le n\}.$$

The prediction at x is its actual conditional correct-label probability $q_S(x) = \mathbb{P}(f(x)G(x) > 0 \mid \mathcal{A}_S)$, and we write

$$F_\theta(S) = \mathbb{P}_x\{q_S(x) \le \theta\}, \qquad \theta = \frac{13}{20}, \qquad c_\theta = \sqrt{2\pi}\left(\frac{1}{e} - \frac{7}{20}\right) > 0.$$

The threshold $\theta < 2/3$ supplies a strict buffer for passage from limiting Gaussian posteriors to finite-width posterior probabilities.

Suppose the Walsh characters are eigenfunctions of the normalized covariance operator, with degree-j eigenvalue λ_j. Write $V_r = \operatorname{span}\{\chi_U : |U| \le r\}$ and $D_r = \dim V_r$. For $r < d$, let Q be the covariance contributed by the complementary Fourier modes, and put

$$\tau = Q(x, x) = \sum_{j > r} \binom{d}{j} \lambda_j, \qquad \ell = \max_{j > r} \lambda_j.$$

The matrix eigenvalues of Q are $M\lambda_j$; this normalization is used throughout. Define $\|h\|_K^2 = h^T K^{-1} h$, and write $u_+ = \max(u, 0)$ and $u_- = \max(-u, 0)$.

Lemma 6.1 (Mean norm and a confidence margin)**.** *Let $m(x) = \mathbb{E}[G(x) \mid \mathcal{A}_S]$, and let $s \le n$ be the number of distinct sampled inputs. Then*

$$\|m\|_K^2 \le s, \qquad f(X_i)m(X_i) > 0.$$

Let $\sigma_x^2 = \operatorname{Var}(G(x) \mid G_S)$, where S here lists the distinct inputs. Whenever $\sigma_x > 0$ and $q_S(x) > \theta$,

$$f(x)m(x) > c_\theta \sigma_x.$$

Proof. Whiten the Gaussian training vector as $G_S = K_{SS}^{1/2}\xi$. The sign constraints form a homogeneous cone in ξ. Conditioning leaves the squared Gaussian radius with mean s, so Jensen's inequality and Gaussian regression give

$$m_S^T K_{SS}^{-1} m_S \le s, \qquad m = K_{\mathcal{X},S} K_{SS}^{-1} m_S.$$

The RKHS norm of this regression vector is the preceding quadratic form. Its observed signed coordinates are strictly positive.

For the confidence claim, the scalar conditional variable $Y = f(x)G(x)$ has a log-concave density: it is a marginal of a Gaussian restricted to a convex set, by the marginal log-concavity theorem [5]. If $\mu = \mathbb{E}Y$ and F_Y is its cdf, log-concavity and Jensen's inequality imply

$$\log F_Y(\mu) \ge \mathbb{E} \log F_Y(Y) = -1.$$

Consequently $\mathbb{P}(Y > \mu) \le 1 - e^{-1} < \theta$. Conditioning first on the full training vector shows that the density of Y is a mixture of Gaussian densities with common variance σ_x^2, and is therefore at most $(\sqrt{2\pi}\sigma_x)^{-1}$. If $q_S(x) > \theta$, then $\mu > 0$ and

$$\theta < \mathbb{P}(Y > 0) \le 1 - e^{-1} + \frac{\mu}{\sqrt{2\pi}\sigma_x}.$$

This gives the stated margin. □

Lemma 6.2 (Most residual variances remain large)**.** *For every training sample,*

$$\mathbb{P}_x\{\sigma_x^2 < \tau/2\} \le \frac{2n\ell}{\tau}.$$

Also, if p is the degree-r Fourier truncation of m and $t = m - p$, then $\|t\|_2^2 \le n\ell$.

Proof. Write $G = G_{\le r} + G_{>r}$ as a sum of independent fields. Conditioning additionally on the full low-degree field can only decrease conditional Gaussian variance. The remaining covariance is

$$Q - Q_{\mathcal{X},S} Q_{SS}^{\dagger} Q_{S,\mathcal{X}}.$$

Its subtracted term is positive semidefinite, bounded above by Q, and has rank at most $s \le n$. Its normalized trace is at most $n\ell$. Markov's inequality proves the first assertion; it includes observed points automatically. Fourier orthogonality and the mean norm bound prove $\|t\|_2^2 \le \ell\|m\|_K^2 \le n\ell$. □

7 Uniform sampling detects the negative part

Lemma 7.1 (Uniform ReLU sampling on the cube)**.** *For every fixed $\delta, \eta \in (0,1)$, there is a universal constant C such that, if*

$$n \ge C\eta^{-2} D_r \log\left(\frac{D_r}{\delta\eta}\right),$$

then with probability at least $1 - \delta$, simultaneously for all $p \in V_r$,

$$\left| \frac{1}{n}\sum_{i=1}^{n} (-f(X_i)p(X_i))_+ - \mathbb{E}_x(-f(x)p(x))_+ \right| \le \eta \mathbb{E}_x|p(x)|.$$

In particular, when $r < k$ and $\eta = 1/4$, the empirical negative part is at least $\mathbb{E}|p|/4$.

Proof. Let A have rows $f(x)(\chi_U(x))_{|U|\le r}$, one for each cube point. Then $A^TA = MI$ and every row has squared norm D_r. The ℓ_1 Lewis weights of A are exactly D_r/M: substitution in their defining equation

$$w_x^2 = A_x(A^T \operatorname{diag}(w^{-1})A)^{-1}A_x^T$$

verifies this constant solution, and Lewis weights are unique. The additive ReLU coreset theorem of Mai, Musco, and Rao [3, Theorem 5] therefore samples precisely the uniform distribution on the cube and gives the stated estimate. Its sample bound is $O(\eta^{-2}D_r \log(D_r/(\delta\eta)))$; no additional logarithmic factor is required. Finally $\mathbb{E}fp = 0$, because $r < k$, so $\mathbb{E}(-fp)_+ = \mathbb{E}|p|/2$. □

8 A uniform bound through the gaps between Fourier levels

We prove a lower bound at each specified sample size using a thin middle band of Fourier modes whose covariance trace is small. Ordinary Lipschitz contraction controls this band, while Lewis-weight sampling controls the lower degrees.

Lemma 8.1 (A fixed-threshold sampling bound)**.** *Let $r < u < d$, and put $\Delta = \tau_r - \tau_u$. Let $\mathcal{B}_{r,u}$ be the span of Walsh characters of degrees $r+1$ through u. Suppose*

$$\eta^2 \ge C\frac{D_r \log(2n)}{n}$$

for a sufficiently large universal constant C. For deterministic $R, T \ge 0$, the class

$$\mathcal{H} = \{-f(p+b) : p \in V_r,\ \mathbb{E}|p| \le R,\ b \in \mathcal{B}_{r,u},\ \|b\|_K \le \sqrt{n}\}$$

satisfies

$$\mathbb{E}_S \sup_{h\in\mathcal{H}} |\mathbb{P}_n(h-T)_+ - \mathbb{E}_x(h-T)_+| \le 4(R\eta + \sqrt{\Delta}). \tag{17}$$

Here $\mathbb{P}_n g = n^{-1}\sum_i g(X_i)$.

Proof. For the full-cube matrix with rows $f(x)(\chi_U(x))_{|U|\le r}$, the Lewis weights are D_r/M. The linear-process moment bound of Mai, Musco, and Rao [3, Theorem 3], with its failure parameter set to $1/2$, therefore implies

$$\mathbb{E}_{S,\epsilon} \sup_{p\in V_r,\,\mathbb{E}|p|\le R} \left|\frac{1}{n}\sum_i \epsilon_i f(X_i)p(X_i)\right| \le R\eta.$$

The normalization follows directly: their sampled matrix rescales each sampled row by M/n, whereas $\|p\|_1 = M^{-1}\|A\beta\|_1$. Their moment bound implies the displayed expectation bound by Jensen.

The covariance-weighted feature vector of the band has squared norm Δ at every input. Consequently

$$\mathbb{E}_{S,\epsilon} \sup_{\|b\|_K\le\sqrt{n}} \left|\frac{1}{n}\sum_i \epsilon_i f(X_i)b(X_i)\right| \le \frac{\sqrt{n}}{n}\sqrt{n\Delta} = \sqrt{\Delta}.$$

The map $z \mapsto (z-T)_+$ is 1-Lipschitz and vanishes at zero. Symmetrization, followed by the scalar contraction principle, bounds the left-hand side of (17) by four times the linear Rademacher width of $\mathcal{H}$. The triangle inequality and the preceding two estimates prove the claim. Both parameter balls and the threshold are fixed before sampling; no positive homogeneity in T or relative bound on an unbounded parameter class is asserted. □

9 A polynomial-threshold obstruction

The following elementary polynomial-threshold obstruction is due to Aspnes, Beigel, Furst, and Rudich [1]; we include its proof.

Lemma 9.1 (Polynomial thresholds cannot approximate parity)**.** *On a cube of dimension k, for $0 \le s < k$, every degree-s polynomial threshold, with a fixed common convention for zero ties, has full-parity error at least*

$$2^{-k} \sum_{j=0}^{\lfloor (k-s-1)/2 \rfloor} \binom{k}{j}. \tag{18}$$

This lower bound tends to $1/2$ whenever $s = o(\sqrt{k})$.

Proof. A sufficiently small constant perturbation resolves any zero ties without changing the specified threshold classifier. Thus suppose P has no zeros. Put $t = \lfloor (k-s-1)/2 \rfloor$. If its error set had fewer than $\sum_{j \le t} \binom{k}{j}$ points, a nonzero polynomial Q of degree at most t could be chosen to vanish on every error. The function fPQ^2 would then be nonnegative everywhere and positive somewhere. But PQ^2, after multilinear reduction, has degree at most $s + 2t < k$, so its parity Fourier coefficient $\mathbb{E} fPQ^2$ is zero, a contradiction. This proves (18). The last assertion follows from the central binomial bound $\max_j 2^{-k}\binom{k}{j} = O(k^{-1/2})$. □

10 Random tail covariance matrices

Let Q be the covariance kernel retaining only degrees greater than an odd cutoff r, and write $\tau = \tau_r$, $\ell = \lambda_{r+1}$, and $s_2 = s_{2,r}$. This section proves the sample event needed later, without assuming that a Gaussian approximation remains valid after conditioning. The entire argument concerns the exact limiting Gaussian covariance.

Lemma 10.1 (Far-pair decay)**.** *There are constants $c_0, C > 0$ such that, when $r \le D/8$ is odd,*

$$\max_{d/4 \le \mathrm{dist}(x,y) \le 3d/4} |Q(x,y)| \le \rho_r := C\sqrt{D}\, e^{-r/8} + C e^{-c_0 D}.$$

Proof. The full-cube Krawtchouk polynomial satisfies

$$\sum_{j=0}^{D} \mathsf{K}_j(h; D) z^j = (1+z)^{D-h}(1-z)^h.$$

If $\delta D \le h \le (1-\delta)D$, put $b = \min(h, D-h)$. On $|z| = t$ the generating function is bounded in absolute value by

$$(1+t^2)^b (1+t)^{D-2b} \le (1+t)^D \exp\left[-\frac{2\delta D t}{(1+t)^2} \right].$$

Cauchy's coefficient bound at $t = j/(D-j)$ and the uniform Stirling lower bound for $\binom{D}{j}$ therefore give, for $1 \le j < D$,

$$\frac{|\mathsf{K}_j(h; D)|}{\binom{D}{j}} \le C\sqrt{D} \exp[-2\delta j(1 - j/D)]. \tag{19}$$

For reduced-cube points at distance in $[d/4, 3d/4]$, their augmented full-cube distance lies in $[D/8, 7D/8]$ for large d. Because r is odd, the exact tail is $Q(x,y) = \sum_{j > r,\ j \text{ odd}} a_j \mathsf{K}_j(h; D)$. For

$r < j \le D - r$, (19) gives a bound $C\sqrt{D}e^{-r/8}$ after summation, since the total spectral mass is one. For $j > D - r$, monotonicity and an odd cutoff b_D near $D/4$ give

$$\sum_{j>D-r} \binom{D}{j} a_j \le a_{b_D} \sum_{k<r} \binom{D}{k} \le CD^{-\beta-1/2} \exp[-D(\mathsf{h}(1/4) - \mathsf{h}(1/8)) + O(1)].$$

Here $\mathsf{h}(t) = -t \log t - (1-t)\log(1-t)$ denotes the binary entropy in nats. The positive entropy gap absorbs the polynomial factor and proves the claim. Adjust constants for bounded d. □

Lemma 10.2 (A tail Gram deviation bound)**.** *Let $X_1, \dots, X_n$ be independent uniform cube points, with $1 \ll n \le M/4$. Let*

$$B_{\text{near}} = \#\{y : \operatorname{dist}(x, y) \notin [d/4, 3d/4]\} \le 2e^{d\mathsf{h}(1/4)}, \qquad q = 2\log M.$$

The count is independent of x. For every $z > 0$,

$$\begin{aligned} &\mathbb{P}\{\|Q_{SS} - \tau I_n\|_{\text{op}} > z\tau\} \\ &\quad \le \frac{C}{z^2}\left[\frac{q\rho_r}{\tau} + \frac{\sqrt{nqs_2}}{\tau} + \frac{n\ell}{\tau} + \frac{n(1+B_{\text{near}})}{M}\right]^2 + \frac{n^2(1+B_{\text{near}})}{M}. \end{aligned} \tag{20}$$

Here the sampled matrix includes repeated training locations.

Proof. Let $J = Q - \tau I_M$, a zero-diagonal Hermitian matrix. Form a deterministic mask J_{far} retaining only pairs at distances in $[d/4, 3d/4]$. Since Q is positive semidefinite with constant diagonal τ, every entry has absolute value at most τ. Thus

$$\begin{aligned} \|J_{\text{far}}\|_{\max} &\le \rho_r, \\ \|J_{\text{far}}\|_{1\to 2}^2 &\le Ms_2, \\ \|J_{\text{far}}\|_{\text{op}} &\le M\ell + \tau + \tau B_{\text{near}}. \end{aligned}$$

For the second inequality, Parseval gives squared column norm $Ms_2 - \tau^2$ for J, and entrywise masking only decreases it. For the third, $\|J\| \le M\ell + \tau$, while the removed matrix has maximum absolute row sum at most τB_{near}. The entropy bound on B_{near} is the elementary binomial tail bound on distances to a fixed cube point.

Let R retain each of the M coordinates independently with probability $p = 2n/M$. The zero-diagonal case of Tropp's random principal-submatrix theorem [4, Theorem 1.1], at moment order two and $q = 2\log M$, states

$$(\mathbb{E}\|RJR\|^2)^{1/2} \le C\left[q(\mathbb{E}\|RJR\|_{\max}^2)^{1/2} + \sqrt{pq}(\mathbb{E}\|JR\|_{1\to 2}^2)^{1/2} + p\|J\|\right].$$

Apply it to J_{far} and substitute the preceding bounds:

$$(\mathbb{E}\|RJ_{\text{far}}R\|^2)^{1/2} \le C[q\rho_r + \sqrt{nqs_2} + n\ell + (n/M)\tau(1 + B_{\text{near}})]. \tag{21}$$

The logarithm here is that of the original dimension M, not the sample size.

To pass to a uniform subset of size n, use a uniform permutation of all cube points, independently of $B \sim \text{Bin}(M, 2n/M)$. Its first B elements form the Bernoulli subset. On $B \ge n$, the first n principal submatrix is a compression of the larger one, so its operator norm cannot be larger. The size-n event is independent of B, and $\mathbb{P}(B \ge n) \ge 1/2$ for large n. Thus Markov's inequality applied to (21) gives the same fixed-size probability bound with a constant factor loss.

Conditional on no collisions, an independent size-n sample is a uniform size-n subset, up to an irrelevant ordering. Collisions have probability at most $n(n-1)/(2M)$. The probability of any sampled pair violating the distance condition is at most $n^2 B_{\text{near}}/M$. On their complement, the sampled matrices J and J_{far} agree. Adding these exceptional probabilities proves (20). □

11 Tail covariance estimates at sparse cutoffs

The original principal-submatrix estimate uses the logarithm of the full cube cardinality. We retain that estimate here. A more precise bound on entries near orthogonal input pairs makes it applicable even when the spectral cutoff is a sufficiently large fixed integer.

Lemma 11.1 (Near-orthogonal entries)**.** *There are constants $a_0, \delta_0, c, C > 0$ such that the following holds. Put $D = d+1$, let r be odd with $r+2 \le \delta_0 D$, and let h be the Hamming distance between the augmented inputs $(1,x),(1,y)$. If $a = |D-2h|/D \le a_0$, then*

$$\frac{|Q_r(x,y)|}{\tau_r} \le [C(a+\sqrt{(r+2)/D})]^{r+2} + Cd^{\beta}e^{-cd}, \qquad Q_r = K_{>r}. \tag{22}$$

Proof. For $1 \le j < D$, the Krawtchouk generating function gives

$$\frac{|\mathsf{K}_j(h;D)|}{\binom{D}{j}} \le [e^{3/2}(a+\sqrt{j/D})]^j. \tag{23}$$

Indeed, set $b = \min(h, D-h)$. On the circle $|z| = t$, the generating function has modulus at most

$$(1+t^2)^b(1+t)^{D-2b} \le \exp(Dt^2/2 + Dat).$$

Cauchy's coefficient bound and $\binom{D}{j} \ge (D/j)^j$, with $t = j/[D(a+\sqrt{j/D})]$, prove (23): both Dt^2 and Dat are at most j.

Choose a_0, δ_0 so small that $e^{3/2}(a_0+\sqrt{\delta_0}) \le e^{-1}$ and $\delta_0 < 1/8$. On $j \le \delta_0 D$, the derivative of $j\log[e^{3/2}(a+\sqrt{j/D})]$ is at most $-1+1/2 < 0$. Consequently (23) is bounded by its value at $r+2$ throughout $r < j \le \delta_0 D$. The sum of the corresponding positive spectral masses $\binom{D}{j}a_j$ is at most τ_r. For $\delta_0 D < j \le (1-\delta_0)D$, the central-distance estimate (19) bounds the normalized Krawtchouk polynomial by $C\sqrt{D}e^{-cD}$; its polynomial factor can be absorbed by decreasing c. For $j > (1-\delta_0)D$, choose an odd b_D near $D/4$. Monotonicity of a_j and Lemma 3.2 bound the remaining mass by

$$a_{b_D} \sum_{i<\delta_0 D} \binom{D}{i} \le \mathrm{poly}(D)e^{-D[\mathsf{h}(1/4)-\mathsf{h}(\delta_0)]},$$

where $\mathsf{h}(t) = -t\log t - (1-t)\log(1-t)$. The entropy gap is positive. Finally $\tau_r \ge cD^{-\beta}$ by Lemma 3.2, proving (22). □

Lemma 11.2 (Sparse adaptive tail Gram matrices)**.** *Fix $A \ge 1$, a failure probability $\delta > 0$, and an odd integer $r_0 \ge 11$. There is $\alpha_0 > 0$ such that, for any fixed $0 < \alpha \le \alpha_0$, the following assertions hold uniformly as $d \to \infty$. Suppose r is odd,*

$$r_0 \le r \le \alpha d, \qquad AD_r\log(D_r+1) \le n < AD_{r+2}\log(D_{r+2}+1).$$

Put $u = r+2$, $\rho = r/(d+1)$, and $H = \log(e/\rho)$. For each such specified n, with probability at least $1-\delta$,

$$\begin{aligned}\|(Q_r)_{SS}\|_{\mathrm{op}} &\le C_{L,\delta}\left[1 + \frac{\sqrt{nd\,s_{2,r}}}{\tau_r} + \frac{n\ell_r}{\tau_r}\right]\tau_r \\ &\le C_{L,\delta}[1+\sqrt{A\rho^{-1}H} + AH]\tau_r, \qquad (24)\\ \|(Q_u)_{SS} - \tau_u I_n\|_{\mathrm{op}} &\le [C_{L,\delta}(\sqrt{A\rho H} + A\rho^2 H) + o(1)]\tau_r. \qquad (25)\end{aligned}$$

The $o(1)$ is uniform over the stated range. In particular, after reducing α_0 if necessary, $\|(Q_u)_{SS}\|_{\mathrm{op}} \le 2\tau_r$ on the same event for all sufficiently large d. Deterministically,

$$0 \le \tau_r - \tau_u \le C_L \tau_r / r, \tag{26}$$

$$n\ell_u/\tau_r \le C_L A \rho^2 H, \tag{27}$$

where $\ell_v = \lambda_{v+1}$. No simultaneous event over different sample sizes is asserted.

Proof. The dimension and spectral estimates already proved give

$$n \le CA\binom{D}{r+2} rH, \qquad \log(2n) \le C_A r H,$$

$$n\ell_r/\tau_r \le C_L A H, \qquad n d\, s_{2,r}/\tau_r^2 \le C_L A \rho^{-1} H.$$

For $u = r+2$, the exact two-step binomial ratio gives

$$n\ell_u/\tau_r \le C_L A \rho^2 H, \qquad n d\, s_{2,u}/\tau_r^2 \le C_L A \rho H.$$

Moreover $\tau_r - \tau_u = \binom{D}{r+2} a_{r+2}$, so the level and trace estimates prove (26).

We apply the proof of Lemma 10.2 with a different deterministic mask. Retain pairs satisfying

$$\left|\frac{1 + x \cdot y}{d+1}\right| \le a_n, \qquad a_n = \sqrt{128 \log(2n)/d}.$$

By choosing α_0 sufficiently small, uniformly $\log(2n)/d \le C_A \alpha \log(e/\alpha) + o(1)$ is as small as needed, so $a_n \le a_0$. For independent uniform inputs, Hoeffding's bound (and $a_n \ge 2/(d+1)$) gives exceptional-pair probability

$$b_n \le 2\exp(-a_n^2 d/8) = 2(2n)^{-16}.$$

The exceptional set is translation invariant on the cube. Its size is therefore the same in every row, so the masked-matrix proof applies with $B_{\mathrm{near}}/M = b_n$. This mask is deterministic for each fixed pair (d, n); it does not depend on the observed sample. Lemma 11.1 supplies the retained-entry bound

$$\frac{\eta_v}{\tau_v} \le \left[C\sqrt{\frac{\log(2n) + v + 2}{d}}\right]^{v+2} + Cd^{\beta} e^{-cd}, \qquad v \in \{r, u\}. \tag{28}$$

Thus exactly the argument of Lemma 10.2, still with $q = 2\log M = 2d\log 2$, gives for every $z > 0$

$$\mathbb{P}\{\|(Q_v)_{SS} - \tau_v I_n\| > z\tau_v\}$$
$$\le \frac{C}{z^2}\left[\frac{q\eta_v}{\tau_v} + \frac{\sqrt{nq s_{2,v}}}{\tau_v} + \frac{n\ell_v}{\tau_v} + n(M^{-1} + b_n)\right]^2 + n^2(M^{-1} + b_n). \tag{29}$$

The collision and exceptional-pair terms tend to zero uniformly: $n \ge AD_{r_0}\log(D_{r_0}+1) \to \infty$, and α_0 can be chosen so that $n \le e^{c_1 d}$ with $2c_1 < \log 2$.

The coherence term $q\eta_v/\tau_v$ also tends to zero uniformly. For $r \le \sqrt{d}$, its bracket in (28) is at most $C_A d^{-1/4}\sqrt{\log d}$, and its exponent is at least $r_0 + 2 \ge 13$. Hence it is bounded, after multiplication by q, by

$$Cd[C_A d^{-1/4}\sqrt{\log d}]^{13} + Cd^{1+\beta}e^{-cd} = o(1).$$

For $r > \sqrt{d}$, reducing α_0 makes the bracket at most $1/2$, so the bound is $Cd\,2^{-\sqrt{d}} + Cd^{1+\beta}e^{-cd} = o(1)$. Using (29) for $v = r, u$, with failure probability at most $\delta/2$ for each, now proves (24) and (25). For the latter, multiply by $\tau_u/\tau_r \le 1$ before using the spectral bounds, so no lower comparison of the two traces is needed. Finally $\rho H \to 0$ as $\rho \to 0$, uniformly for $0 < \rho \le \alpha$; choose α_0 to make the fixed part of the fine deviation below $1/2$, and absorb the uniform $o(1)$. Since $\tau_u \le \tau_r$, this gives the asserted upper bound $2\tau_r$. □

12 A lower bound for sparse parity at each specified sample size

Let $U \subseteq [d]$, $|U| = k$, and $f = \chi_U$. Write $F_\theta(S) = \mathbb{P}_x\{q_S(x) \le \theta\}$ for the actual Gaussian sign-posterior confidence failure, with

$$\theta = \frac{13}{20}, \qquad c_\theta = \sqrt{2\pi}\left(\frac{1}{e} - \frac{7}{20}\right), \qquad a_\theta = \frac{c_\theta}{2\sqrt{2}}.$$

The next theorem has a failure fraction independent of both k and d.

Theorem 12.1 (Ambient-dimension lower bound)**.** *For each fixed depth $L \ge 2$, there are constants $c > 0$ and d_0 such that, whenever $d \ge d_0$, $1 \le k \le d$, and*

$$0 \le n \le \exp\left(ck \log \frac{ed}{k}\right),$$

independent uniform training inputs satisfy

$$\mathbb{P}_S\{F_{13/20}(S) \ge \varepsilon\} \ge \frac{3}{4}, \qquad \varepsilon = \frac{c_\theta^2}{256} > 0.$$

The assertion holds for each specified sample size. In the network model of the width-limit proposition, every fixed $\gamma < 0$ consequently satisfies the same lower bound for $\liminf_{N\to\infty} \mathbb{P}_x\{q_{N,S}(x) \le 2/3\}$.

Proof. Fix a failure probability $\delta = 1/40$. We choose a sampling constant A sufficiently large, a degree fraction $\alpha > 0$ sufficiently small, and an odd integer $r_0 \ge 11$ sufficiently large, in that order. All these choices depend only on the fixed depth and universal constants. Further decreases in α are harmless.

We first treat $k \ge K_0$, where the fixed constant K_0 is chosen below; bounded k will be covered at the end. For a specified $n > 0$, let r be the largest odd integer satisfying

$$AD_r \log(D_r + 1) \le n. \tag{30}$$

First suppose this cutoff exists and $r \ge r_0$. By choosing c in the theorem sufficiently small and then K_0 sufficiently large, we ensure

$$r \le \alpha k \le \alpha d, \qquad u := r + 2 < k, \qquad n < AD_{r+2} \log(D_{r+2} + 1). \tag{31}$$

Indeed, the binomial lower bound $\binom{d}{r} \ge (d/r)^r$ shows that, for $r \ge \alpha k$, the logarithm of the sampling threshold is at least a constant multiple of $\alpha k \log(ed/k)$. Rounding by two can be absorbed by increasing K_0. Also $k \log(ed/k) \le d$, so the same choice of c puts n within the small-exponential range needed for the random Gram bounds. Uniformly in this shell interval, for large d,

$$\log(2n) \le C \log(D_r + 1) \le Cr \log(eD/r), \qquad D = d + 1.$$

Put $\rho = r/D$, $H_\rho = \log(e/\rho)$, $\tau = \tau_r$, $\ell_r = \lambda_{r+1}$, and

$$B = C_{L,\delta}\left(1 + \frac{\sqrt{nd\, s_{2,r}}}{\tau} + \frac{n\ell_r}{\tau}\right), \qquad \eta^2 = C_0 \frac{D_r \log(2n)}{n}.$$

Choose C_0 as required by Lemma 8.1. Lemma 11.2 and its near-orthogonal coherence estimate imply, with probability at least $1 - \delta$ for all sufficiently large d,

$$\|Q_{r,SS}\|_{\mathrm{op}} \le B\tau. \tag{32}$$

This holds uniformly over $r \geq r_0$ in (31); it does not require $r \gg \log d$. The sharp spectral ratios give the crucial cancellation

$$\eta^2 B \leq C_{L,\delta} \left[\frac{1}{A} + \sqrt{\frac{\rho H_\rho}{A}} + \rho^2 H_\rho \right]. \tag{33}$$

For completeness, the constant term uses $\eta^2 \leq C/A$. The eigenvalue term is bounded by

$$\eta^2 \frac{n\ell_r}{\tau} \leq \frac{C D_r \log(2n)}{r\binom{D}{r+2}} \leq C\rho^2 H_\rho.$$

The square of the variance term is at most

$$\left(\eta^2 \frac{\sqrt{nd\, s_{2,r}}}{\tau} \right)^2 \leq \frac{C}{A} \frac{C D_r \log(2n) d}{r^2 \binom{D}{r+2}} \leq \frac{C}{A} \rho H_\rho.$$

Thus $\eta\sqrt{B}$ can be made arbitrarily small uniformly in n by choosing A large and then α small.

Let $m = \mathbb{E}[G \mid \mathcal{A}_S]$ and $p = \Pi_{\leq r} m$. By Lemma 6.1, $\|m\|_K^2 \leq n$ and its training signs are correct. The ordinary first-power ReLU sampling event in Lemma 7.1, at accuracy $1/4$, has probability at least $1 - \delta$ after increasing A. Since $r < k$, Walsh orthogonality gives $\mathbb{E} fp = 0$, so on this event and (32),

$$\frac{1}{4}\mathbb{E}|p| \leq \mathbb{P}_n(-fp)_+ \leq \sqrt{\mathbb{P}_n(m-p)^2} \leq \sqrt{B\tau}. \tag{34}$$

Set the deterministic radius $R = 4\sqrt{B\tau}$.

Only one additional paired level is required. Write

$$m = p + b + t, \qquad b = \Pi_{(r,u]} m, \quad t = \Pi_{>u} m, \quad u = r+2, \qquad \Delta = \tau_r - \tau_u.$$

The level-mass estimate and the two-step binomial ratio give

$$\Delta/\tau \leq C_L/r, \tag{35}$$

$$n\ell_u/\tau \leq C_L A \rho^2 H_\rho, \tag{36}$$

$$nd\, s_{2,u}/\tau^2 \leq C_L A \rho H_\rho. \tag{37}$$

For example, maximality yields $n \leq CA\binom{D}{r+2} r H_\rho$, whereas the two high-tail bounds involve $\binom{D}{r+4}$ in the denominator; its ratio with $\binom{D}{r+2}$ supplies a factor $O(\rho^2)$. By increasing r_0 we have $\tau_u \geq 3\tau/4$. By reducing α, Lemma 11.2 applied at u now gives, with probability at least $1 - \delta$,

$$\|Q_{u,SS}\|_{\mathrm{op}} \leq 2\tau. \tag{38}$$

The coherence remainder tends to zero uniformly, including at fixed r_0. The two displayed high-tail terms can be made as small as needed before taking d large.

Put $\zeta = n\ell_u/\tau$. We choose α small enough that

$$e_* := (4 + a_\theta^{-2})\zeta \leq a_\theta^2/32. \tag{39}$$

This is possible uniformly by (36). The RKHS bound, the high-tail Gram event, and the variance rank bound imply

$$\mathbb{P}_n t^2 \leq 2\tau, \qquad \|t\|_2^2 \leq \zeta\tau, \qquad \mathbb{P}_x\{\sigma_x^2 < \tau/2\} \leq 4\zeta.$$

For the last assertion, condition additionally on the full degree-u field. Its remaining diagonal before training is $\tau_u \geq 3\tau/4$, and the normalized trace of the variance reduction is at most $n\ell_u = \zeta\tau$. Consequently, except on an input set of measure at most e_*, both $\sigma_x^2 \geq \tau/2$ and $|t(x)| \leq a_\theta\sqrt{\tau}$ hold. The confidence lemma then shows that

$$f(x)(p(x) + b(x)) > a_\theta\sqrt{\tau} \tag{40}$$

at every remaining point where $q_S(x) > \theta$.

A fixed stop-loss bound rules out hiding the negative mass. Choose the deterministic threshold $T = 4\sqrt{\tau}/a_\theta$. Apply Lemma 8.1 to the deterministic radius $R = 4\sqrt{B\tau}$ and this T. By Markov's inequality its uniform empirical error is at most

$$4\delta^{-1}(R\eta + \sqrt{\Delta}) \leq a_\theta\sqrt{\tau}/8$$

with probability at least $1 - \delta$, after the stated choices: (33) controls the first term and (35) controls the second by increasing r_0. The actual p, b belong to this class on the earlier events, since (34) bounds $\mathbb{E}|p|$ and $\|b\|_K \leq \sqrt{n}$. Let $h = p + b$. Training consistency gives $(-f_i h_i)_+ \leq |t_i|$. Using $(z - T)_+ \leq z^2/(4T)$ for $z \geq 0$ and (38), we obtain

$$\mathbb{P}_n(-fh - T)_+ \leq \frac{\mathbb{P}_n t^2}{4T} \leq a_\theta\sqrt{\tau}/8, \qquad \mathbb{E}_x(-fh - T)_+ \leq a_\theta\sqrt{\tau}/4.$$

By (40), the positive-part integral of fh is at least $a_\theta\sqrt{\tau}(1 - F_\theta - e_*)$, and its negative set has measure at most $F_\theta + e_*$. Crucially, $\deg h \leq u < k$, so $\mathbb{E}fh = 0$ exactly. Equality of the positive and negative integrals therefore gives

$$a_\theta(1 - F_\theta - e_*) \leq \frac{4}{a_\theta}(F_\theta + e_*) + \frac{a_\theta}{4}.$$

It follows that

$$F_\theta \geq \frac{3a_\theta^2}{4(4 + a_\theta^2)} - e_* \geq \frac{a_\theta^2}{32} = \varepsilon,$$

where we used (39) and $a_\theta^2 < 1$. The low Gram, high Gram, ReLU, and fixed-stop-loss events have combined probability at least $1 - 4\delta > 3/4$ for all sufficiently large d. No independence between these events is assumed.

The remaining, smaller samples. If the maximal cutoff is below r_0, or does not exist, then $n < AD_{r_0}\log(D_{r_0} + 1)$. Fix the odd integer $s = r_0 + 2$. The sharp spectrum gives, uniformly in this entire range,

$$\frac{n\ell_s}{\tau_s} \leq C_{L,A,r_0}\frac{D_{r_0}\log(D_{r_0} + 1)}{s\binom{D}{s+2}} = o(1) \qquad (d \to \infty).$$

Let $p = \Pi_{\leq s}m$. The mean-norm, residual-variance, and confidence lemmas show that, except on an $o(1)$ fraction of inputs, confidence above θ implies $fp > 0$. Conditional on any values of the $d - k$ nuisance coordinates, p is a degree-s polynomial on the k target coordinates. Lemma 9.1 thus bounds its strictly correct fraction by $1 - b_k(s)$, where

$$b_k(s) = 2^{-k}\sum_{j=0}^{\lfloor (k-s-1)/2\rfloor}\binom{k}{j}.$$

Choose K_0 sufficiently large relative to the fixed s that $b_k(s) \geq 2/5$ for every $k \geq K_0$. This gives $F_\theta \geq 2/5 - o(1) > \varepsilon$ for every such training sample. At $n = 0$, $q_S = 1/2$ identically. All errors above are uniform over $k \geq K_0$ and the specified sample range after fixing the constants.

Finally shrink c, if necessary, so that $c \leq 1/(2K_0)$. For $1 \leq k < K_0$ the sample bound implies $n \leq \sqrt{ed}$. The degree-zero projection $p_0 = \Pi_{\leq 0} m$ is constant. The first paired eigenvalues satisfy $\lambda_0 = \lambda_1$; since the total trace is one, $(d+1)\lambda_0 \leq 1$. Consequently its tail satisfies the exact bound $\ell_0/\tau_0 = \lambda_1/(1-\lambda_0) \leq 1/d$. The same mean and residual-variance estimates show that, outside a fraction $O_\theta(n/d) = o(1)$ of inputs, confidence above θ implies $fp_0 > 0$. A constant can be strictly correct on at most half the cube for any nonconstant parity. Therefore $F_\theta \geq 1/2 - o(1) > \varepsilon$ for every training sample in this remaining range. A single sufficiently large d_0 covers all cases.

Finally, the Gaussian sign-posterior width limit holds at fixed d, k, S for every fixed $\gamma < 0$. At any point with $q_S(x) \leq 13/20$, finite-width correct-label probabilities are eventually below $2/3$. Since the cube is finite, the asserted $\liminf$ transfer holds simultaneously at all these points. □

13 Scope of the conclusions

Theorem 12.1, together with Proposition 2.2, proves Theorem 1.1. Theorems 4.2 and 4.3 prove Theorem 1.2. Parity has a Boolean circuit with $O(k)$ gates, whereas the negative exponent posterior requires the stated number of examples to make its confidence failure vanish. This is a property of the specified posterior; parity can be identified from $O(d)$ samples by linear algebra over the two-element field. Searching the supports of size k instead requires only $O(k \log(ed/k))$ samples at fixed success probability, since each incorrect parity agrees with the target independently with probability $1/2$ on each example.

Two questions remain outside the result. At $\gamma = 0$, conditioning retains a fixed margin at all 2^d inputs, so the homogeneous-cone argument used here is unavailable. For $0 < \gamma < 1$, the posterior selects energy minimizers, and the dictator example shows that its learning behavior need not match the sign-conditioned Gaussian regime. Neither the proof nor the counterexample decides whether the same $\exp\{ck \log(ed/k)\}$ lower bound holds for growing sparse k in either of these two regimes.

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